

Elements of combinatorics

— counting by listing

Before any formula, learn to list without repeating and without missing anything.

LESSON 83–85

3 × 40 MIN

ALGEBRA 8, CHAPTER IV §27

STAGE 9 · 12.1

LESSON CLOCK

6'

12'

10'

8'

4'

Warm-up: how many two-digit numbers use only 1, 2, 3?

6 min

Systematic listing

12 min

Tables of outcomes

10 min

Tree diagrams

8 min

Homework

4 min

LEARNING OBJECTIVES

- List every outcome of a small experiment in a systematic order.
- Use a table of outcomes for two stages.
- Draw a tree diagram and read the outcomes off its branches.
- Decide whether order matters in a given counting question.

TEXTBOOK

National

Chapter IV, pp. 181–188

Cambridge

Learner’s Book pp. 255–260

Workbook

Workbook 12.1

01

Explanation

Listing in an order you can trust

Combinatorics is the art of counting without counting one by one. But every method starts from the same discipline: **list in a fixed order**, so that nothing is written twice and nothing is forgotten.

How many two-digit numbers can be made from the digits 1, 2, 3 if a digit may repeat? Fix the first digit, then run through the second:

11, 12, 13 | 21, 22, 23 | 31, 32, 33

Nine numbers — three blocks of three. The blocks are what make the list trustworthy: as soon as you are working alphabetically, or from smallest to largest, a missing item is visible.

THE RULE OF THE LIST

Choose one thing to vary **last**. Hold everything else still while it runs through all its values, then step the next thing on by one.

Does order matter?

Two questions that sound the same are not the same:

QUESTION	ORDER?	FROM {A, B, C} CHOOSE 2	HOW MANY
Who is captain and who is vice-captain?	matters	<i>AB, BA, AC, CA, BC, CB</i>	6
Which two go on the trip?	does not matter	<i>AB, AC, BC</i>	3

When order does not matter, *AB* and *BA* are one and the same choice, so the count is halved. Always ask the question in words before you start listing.

ALSO ASK: MAY THINGS REPEAT?

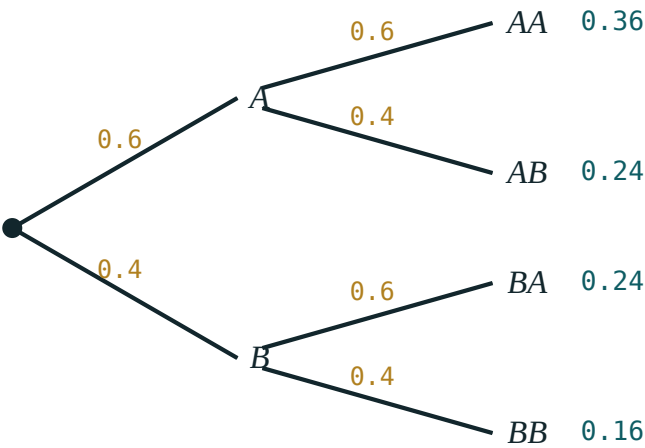
Two dice can both show 4 (repetition allowed). Two different pupils cannot be the same pupil (repetition not allowed). The answer changes the list.

Tables and tree diagrams

For an experiment in **two** stages, a table lists every outcome at once. Two dice:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Thirty-six squares, so thirty-six equally likely outcomes. A sum of 7 fills a whole diagonal — six squares — which is why 7 is the commonest total.



multiply along a branch · add down the ends

A tree diagram. Each path from the root to a tip is one outcome; count the tips to count the outcomes.

A **tree diagram** handles three stages and more, and it keeps the order visible. Read one outcome by walking a single path from left to right.

02 Worked examples

EXAMPLE
1

A café offers 3 soups and 4 main dishes. List, systematically, the ways to choose one soup and one main.

1

$S_1M_1, S_1M_2, S_1M_3, S_1M_4$
Hold the soup still; run the main through all four values.

2

$S_2M_1, S_2M_2, S_2M_3, S_2M_4$
Step the soup on by one.

3

$S_3M_1, S_3M_2, S_3M_3, S_3M_4$
And again.

4

$3 \text{ blocks} \times 4 = 12$
The blocks are equal, so the total is a product.

ANSWER

12 meals

EXAMPLE 2 From the letters A, B, C, D, how many two-letter “words” can be made with no letter repeated?

① *AB, AC, AD*

First letter A — three choices left.

② *BA, BC, BD*

First letter B.

③ *CA, CB, CD, DA, DB, DC*

C and D likewise.

④ $4 \times 3 = 12$

Four first letters, three second letters each.

ANSWER

12 words

EXAMPLE 3 A coin is tossed three times. List all the outcomes and count those with exactly two heads.

① *HHH, HHT, HTH, HTT*

First toss H; the other two run through all four possibilities.

② *THH, THT, TTH, TTT*

First toss T.

③ *8 outcomes in total*

$2 \times 2 \times 2 = 8$ — three stages, two ways each.

④ *HHT, HTH, THH*

Pick out the ones with exactly two H.

ANSWER

8 outcomes; 3 of them have exactly two heads

03 Interactive model

Reveal one counting task at a time and let the class finish the list before you show the next step.

Count it by listing

Digits 1, 2, 3, 4. Two-digit numbers, digits **different**.

① 12, 13, 14

First
digit
1.

② 21, 23, 24

First
digit
2.

③ 31, 32, 34

First
digit
3.

④ 41, 42, 43

First
digit
4 —
four
blocks
of
three.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Outcome	Natija	Исход
Sample space	Elementar natijalar fazosi	Пространство исходов
Systematic listing	Tartibli sanab chiqish	Систематический перебор
Tree diagram	Daraxt diagrammasi	Дерево вариантов
Table of outcomes	Natijalar jadvali	Таблица исходов
Arrangement (order matters)	O'rin almashtirish	Размещение (порядок важен)
Selection (order does not matter)	Tanlash	Сочетание (порядок не важен)
Repetition allowed	Takrorlanish bilan	С повторением
Exhaustive list	To'liq ro'yxat	Полный перечень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

How many two-digit numbers use only the digits 5, 6, 7, repeats allowed?

From 5 people, how many ways can a captain and a vice-captain be chosen?

Two dice are thrown. How many outcomes are there altogether?

In which question does order NOT matter?

Who wins gold and who wins silver?

Which two books do I borrow?

What is the PIN code?

Who sits first and who sits second?

A tree diagram for three stages with 2 branches each has how many tips?

6

8

9

12

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *List all two-digit numbers formed from 1, 2 with repeats allowed.*

ANSWER 11, 12, 21, 22

2. *List all two-letter words from A, B, C with no repeats.*

ANSWER AB, AC, BA, BC, CA, CB

3. *How many outcomes when a coin is tossed twice?*

ANSWER 4

4. *List the ways to choose 2 fruits from apple, pear, plum.*

ANSWER apple–pear, apple–plum, pear–plum

5. *How many outcomes when a die is thrown and a coin tossed?*

ANSWER 12

6. *List all three-digit numbers using each of 1, 2, 3 once.*

ANSWER 123, 132, 213, 231, 312, 321

7. *Does order matter in “choose 3 pupils for the choir”?*

ANSWER No.

Medium · the standard question

7 PROBLEMS

1. *How many two-digit numbers from 0, 1, 2, 3 (no repeats, no leading 0)?*

ANSWER $3 \times 3 = 9$

2. *A menu has 2 starters, 3 mains, 2 desserts. How many meals?*

ANSWER 12

3. *Two dice are thrown. How many outcomes give a total of 8?*

ANSWER 5 — (2,6),(3,5),(4,4),(5,3),(6,2)

4. *From 6 people, how many ways to pick a president and a secretary?*

ANSWER $6 \times 5 = 30$

5. *From 6 people, how many ways to pick a committee of 2?*

ANSWER $30 \div 2 = 15$

6. *A coin is tossed four times. How many outcomes?*

ANSWER $2^4 = 16$

7. *How many outcomes of two dice have both numbers the same?*

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. *How many three-digit numbers use only 1, 2, 3 and contain at least one 3?*

ANSWER $27 - 8 = 19$

2. *How many two-digit even numbers can be made from 1, 2, 3, 4, 5 with no repeats?*

ANSWER $2 \times 4 = 8$

3. *Two dice are thrown. In how many outcomes is the total a prime number?*

ANSWER 15 — totals 2, 3, 5, 7, 11 give $1 + 2 + 4 + 6 + 2$

4. *From 5 pupils, how many ways to seat 3 of them in a row?*

ANSWER $5 \times 4 \times 3 = 60$

5. *How many four-letter “words” from A, B with repeats, having exactly two A’s?*

ANSWER 6 — AABB, ABAB, ABBA, BAAB, BABA, BBAA

6. *A PIN uses 4 digits from 0–9, repeats allowed. How many begin with an odd digit?*

ANSWER $5 \times 10^3 = 5000$

7. *Explain why choosing 2 from n gives half of $n(n - 1)$.*

ANSWER Each unordered pair is counted twice among the $n(n - 1)$ ordered pairs — once as XY and once as YX .

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 181–188. Show your list, not just the number.

1. List all two-digit numbers made from 2, 4, 6 with repeats allowed.
2. From A, B, C, D choose 2 with order mattering — list them all.
3. A shop sells 4 shirts and 3 ties. How many shirt-and-tie sets?
4. Two dice are thrown. List the outcomes with a total of 9.
5. A coin is tossed three times. List the outcomes with at most one head.

6. Write one counting question where order matters and one where it does not.